Matrices, Correlations, and Data Structures

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Why Data Science?

Every day we generate data:

- Social media posts
- YouTube views
- GPS locations
- Online purchases
- Weather measurements

Question: How can computers organize and learn from all this information?
Answer:

1. Data Structures
2. Matrices
3. Correlation Matrices

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What is Data?

Data is information.

Student	Age	GPA
Alice	16	3.9
Bob	17	3.7
Carol	16	4.0

Questions:

- Which student has highest GPA?
- What is the average GPA?
- Is age related to GPA?

To answer these questions efficiently, computers need good data structures.

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Data Structures

A data structure is a way of organizing information.

Examples:

- Lists
- Arrays
- Tables
- Trees
- Graphs

Think of a data structure as a special container for information.

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Everyday Example

Imagine your contacts app. Each contact contains:

- Name
- Phone Number
- Email Address

The phone stores contacts using data structures so information can be found quickly.

Arrays

Example: Scores $=$ [92, 85, 98, 77, 88]
Questions:

- Largest score?
- Average score?
- Median score?

Arrays are one of the most important data structures in data science.

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From Arrays to Matrices

What if we have many variables?

Student	Math	Science
Alice	90	95
Bob	80	85
Carol	100	98

This becomes a matrix.

What is a Matrix? A matrix is a rectangular arrangement of numbers.

Example: $A = \begin{bmatrix} 90 & 95 \\ 80 & 85 \\ 100 & 98 \end{bmatrix}$

- Rows = students
- Columns = subjects

Matrices are everywhere in data science.

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Why Matrices Matter

- Images:
  - A grayscale image is a matrix of pixel values.
- Recommendation Systems:
  - Rows = users
  - Columns = movies
- Weather Forecasting:
  - Rows = locations
  - Columns = measurements

Data science loves matrices.

Matrix Operations: Addition

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 1 \end{bmatrix}, \ B = \begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 1 & 3 \end{bmatrix}$$

Adding corresponding entries

$$A + B = \begin{bmatrix} 3 & 3 \\ 5 & 5 \\ 4 & 4 \end{bmatrix}$$

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Matrix Operations: Multiplication

One of the most important operations in AI

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}, \; B = \begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 1 & 3 \end{bmatrix}$$

$$A * B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 11 & 14 \\ 14 & 11 \end{bmatrix}$$

Example: Student Scores $\times$ Weights. Produces final course grades.

- Matrix multiplication combines information.
- Google, ChatGPT, and machine learning systems use billions of matrix multiplications every second.

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Student	Homework	Exam
Alice	90	95
Bob	80	85
Carol	100	98

Weights:
Homework $= 40\%$
Exam $= 60\%$
Matrix multiplication computes final grades automatically.

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Finding Relationships

Suppose we collect:

- Hours studied
- GPA

Question: Do students who study more tend to have higher GPAs?
This leads to correlation.

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Correlation

Correlation measures how strongly two variables move together.
Values range from: $-1 \leq$ Correlation ≤ 1

- $1 =$ strong positive relationship
- $0 =$ no linear relationship
- $-1 =$ strong negative relationship

Hours	2	3	4	5	6	7	8
GPA	2.55	2.7	3.0	3.20	3.4	3.6	3.75

Table: Hours Studied vs GPA

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Correlation: Example

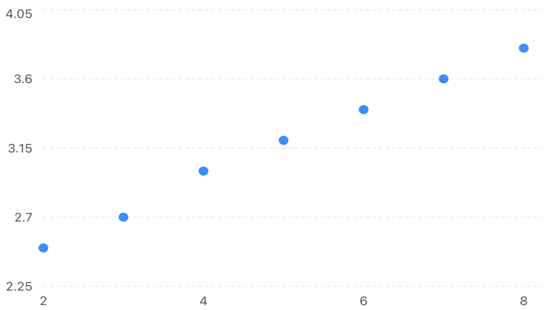

Figure: Hours Studied vs GPA

Question: What do you predict the correlation will be?
As hours increase, GPA increases. Positive correlation.

Correlation Matrix

Suppose we measure

- Height, Weight, GPA

Student	Height (in)	Weight (lb)	GPA
A	60	100	3.0
B	64	115	3.2
C	68	130	3.8
D	70	145	3.4
E	74	160	3.6

Table: Student data for height, weight, and GPA

- $r = \frac{\sum(X_i - \overline{X})(Y_i - \overline{Y})}{\sqrt{\sum(X_i - \overline{X})^2 \sum(Y_i - \overline{Y})^2}}$

Correlation Matrix

	Height	Weight	GPA
Height	1	0.99	0.78
Weight	0.99	1	0.71
GPA	0.78	0.71	1

This summarizes all pairwise relationships at once.

- Height vs Height $= 1$, Weight vs Weight $= 1$, and GPA vs GPA $= 1$
  - Each variable is perfectly correlated with itself
- Height vs Weight: 0.99 (very strong positive)
  - As height increases, weight almost always increases.
  - The relationship is nearly linear.
- Height vs GPA: 0.78 (moderately strong positive)
  - Taller students tend to have slightly higher GPA.
  - But the relationship is not perfect.
  - This does NOT mean height causes GPA to increase. They vary together in this dataset.

Why Correlation Matrices Matter

Used in:

- Machine Learning
- Finance
- Medicine
- Sports Analytics
- Social Networks

They help identify patterns hidden in data.

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Real Data Science Example

Netflix Recommendation System

- Rows $=$ Users
- Columns $=$ Movies
- Entries $=$ Ratings
- The system learns patterns from this matrix.

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Conclusion

Data Structures: Organize information efficiently.

Matrices: Store and manipulate data.

Correlation Matrices: Reveal relationships among variables.

These ideas form part of the mathematical foundation of modern data science.

Thank You! Questions?

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